THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 18	There are $2n = 18$ numbers between n^2 and $(n+1)^2$
2	**(d)** 8	$\frac{1}{a^{-n}} = a^n = 8$
3	**(c)** IC	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
4	**(b)** 144°	One part $= 360 \div 10 = 36^\circ$
5	**(b)** 3465	alternate digit sums 9 and 9 differ by 0, a multiple of 11
6	**(c)** $x^2 + 6x + 9$	$(a+b)^2 = a^2 + 2ab + b^2$
7	**(b)** 4 : 1	Convert to the same unit
8	**(b)** 25%	Change/original × 100 = 20/80×100
9	**(c)** $\sqrt{74}$	$h^2 = 5^2 + 7^2 = 74$
10	**(d)** 8:12:15	Make B equal: B $= \mathrm{LCM}(3,4) = 12$; A $= 8$, C $= 15$ gives $8 : 12 : 15$
11	**(a)** 8	$F = 2 + E - V = 8$
12	**(a)** 16	Even number of values: mean of the two middle values
13	**(d)** $5n$	$5(n+3) - 15 = 5n$
14	**(a)** 15	$b = A/l = 360/24$
15	**(b)** 29	$(n+1)^2 - n^2 = 2n + 1 = 29$
16	**(d)** 81	Even power gives a positive value; odd power gives a negative value
17	**(b)** 73	Multiply each digit by the power of the base and add
18	**(c)** 30	Opposite angles are equal, so the two expressions are equal
19	**(b)** 4518	even and digit sum 18 is a multiple of 3
20	**(b)** $x^2 - 4x + 4$	$(a-b)^2 = a^2 - 2ab + b^2$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 100$ (2).	2
22	$(2^3)^4 = 2^{12}$ (1); $2^{12} \times 2^{-5} = 2^7$ (1).	1+1
23	$3 \times 5^2 + 1 \times 5^1 + 4 \times 5^0$ (1); $= 84$ (1).	1+1
24	Total parts 12, one part $= 360 \div 12 = 30^\circ$ (1); smallest angle $= 30^\circ$ (1).	1+1
25	Sum of digits at odd places and even places: 13 and 13 (1); difference 0, so yes (1).	1+1
26	$x^2 + 2 \times x \times 8 + 8^2$ (1) $= x^2 + 16x + 64$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$T_5 = 15$, $T_6 = 21$ (1+1); sum $= 36 = 6^2$ (1).	1+1+1
28	(i) 2^3 (1); (ii) 2^{-6} (1); (iii) 2^7 (1).	1+1+1
29	$137 = 5 \times 27 + 2$; $27 = 5 \times 5 + 2$; $5 = 5 \times 1 + 0$; $1 = 5 \times 0 + 1$ (1); 1022_5 (1); check $1 \times 5^3 + 0 \times 5^2 + 2 \times 5^1 + 2 \times 5^0 = 137$ (1).	1+1+1
30	Let the angles be $x, 2x, 3x, 4x$; sum $= 10x = 360$ (1); $x = 36$ (1); angles 36°, 72°, 108°, 144° (1).	1+1+1
31	By 8: Yes, last three digits 640 are divisible by 8 (1); by 11: Yes, alternate digit sums 9 and 9 differ by 0, a multiple of 11 (1); conclusion (1).	1+1+1
	OR: $(2x)^2 + 2(2x)(3) + 3^2$ (2); $= 4x^2 + 12x + 9$ (1).	2+1
32	Make B equal: multiply first ratio by 2, second by 1 (1); A : B : C $= 4 : 6 : 5$ (2).	1+2
	OR: GST = 18% of 45000 = ₹ 8100 (2); bill = 45000 + 8100 = ₹ 53100 (1).	2+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$1^3 + 12^3 = 1 + 1728 = 1729$ (2); $9^3 + 10^3 = 729 + 1000 = 1729$ (2); two different ways, so it is a taxicab number (1).	2+2+1
34	(i) $x + 3 = 11$, $x = 8$ (2); (ii) $4x = 20$, $x = 5$ (2); (iii) $x - 2 = -3$, $x = -1$ (1).	2+2+1
35	Column addition with carries gives 10100_2 (3); decimal check: $13 + 7 = 20$ and $10100_2 = 20$ (2).	3+2
36	(i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (78 + 92 + 104) = 86^\circ$ (2).	3+2
	OR: Conditions of 5 and 9 on the stars (2); numbers: 4230, 4635 (2); verify one (1).	2+2+1
37	Area $= (x + 3)(x + 5)$ (1); $= x^2 + 8x + 15$ (2); at $x = 4$: 63 cm^2 (2).	1+2+2
	OR: One part ₹ 150 (1); shares ₹ 150, ₹ 450, ₹ 600 (2); after the transfer ₹ 250, ₹ 450, ₹ 500 (1); new ratio $5 : 9 : 10$ (1).	1+2+1+1
38	A = P(1+R/100)³ = ₹ 10648 (2); CI = ₹ 2648 (1); SI = 8000×10×3/100 = ₹ 2400 (1); difference = ₹ 248 (1).	2+1+1+1
	OR: Half-diagonals 20 cm, 21 cm (1); side$^2 = 400 + 441 = 841$, side $= 29$ cm (2); perimeter $= 4 \times 29 = 116$ cm (1); area $= \frac{1}{2} \times 40 \times 42 = 840$ cm² (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 4 : 6 : 5	Make the middle terms equal
2	**(a)** 25%	18/72×100
3	**(a)** 130	Square on hypotenuse $= a^2 + b^2 = 130$
4	**(a)** 15	$18 \times 10 = 12 \times x$
5	**(b)** 4	Each segment becomes 4: $4^1 = 4$
6	**(d)** 20	Total − known sum
7	**(b)** $3n + 3$	$n + (n+1) + \cdots = 3n + 3$
8	**(d)** 9	$h = 2A/b = 2 \times 72/16$
9	**(b)** 36	$T_n = n(n+1)/2 = 36$
10	**(a)** 6×10^7	Place the decimal point after the first non-zero digit
11	**(d)** 322	Divide repeatedly by the base and read the remainders upwards
12	**(b)** 35°	A diagonal of a rhombus bisects the angle at the vertex
13	**(c)** 6174	Kaprekar routine: arrange digits in descending and ascending order and subtract
14	**(d)** $x^2 - 9$	$(a+b)(a-b) = a^2 - b^2$
15	**(c)** 17.5 km	3.5×5
16	**(b)** 20%	(SP−CP)/CP×100 = 300/1500×100
17	**(c)** 16, 63, 65	$2m = 16$, $m^2 - 1 = 63$, $m^2 + 1 = 65$
18	**(a)** 9	$135 \div 15$
19	**(a)** $\frac{9}{16}$	Each stage keeps $\frac{3}{4}$: $(3/4)^2$
20	**(c)** 22	New sum 132, new count 6

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	3 m 60 cm $= 360$ cm (1); ratio $= 4 : 1$ (1).	1+1
22	$36/90 \times 100$ (1) $= 40\%$ (1).	1+1
23	Take $2m = 28$, so $m = 14$ (1); triplet $(28, 195, 197)$ (1).	1+1
24	$\frac{5}{9} = 0.56$ and $\frac{3}{5} = 0.6$ (1); so 3:5 is greater.	1+1
25	Edges $= 3n = 21$ so $n = 7$ (1); faces $= n + 2 = 9$ (1).	1+1
26	Ascending: 12, 21, 34, 38, 44, 45 (1); mean of the two middle values $= 36$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$2x^2 = (4\sqrt{2})^2 = 32$ (1); $x = 4$ cm (1); area $= \frac{1}{2} \times 4 \times 4 = 8$ cm² (1).	1+1+1
28	Length $= 14$ km (1); breadth $= 8$ km (1); perimeter $= 2(14 + 8) = 44$ km (1).	1+1+1
29	(i) $12(n-2) = 12 \times 1 = 12$ (1+1); (ii) $6(n-2)^2 = 6 \times 1 = 6$ (1).	2+1
30	Mean $= 61$ (1); median $= 66.5$ (1); range $= 58$ (1).	1+1+1
31	Row 2: 5, 4, 5 (1); row 3: 9, 9 (1); top $= 18$ (1).	1+1+1
	OR: Area $= 32 \times 18 = 576$ m² (1); cost $= 576 \times 12$ (1) $=$ ₹ 6912 (1).	1+1+1
32	$22^2 = 484$, $120^2 = 14400$ (1); sum $= 14884$ (1); $122^2 = 14884$ equals the sum, so it is a triplet (1).	1+1+1
	OR: Add the exponents: $\left(\frac{2}{3}\right)^{-2}$ (1); $= \left(\frac{3}{2}\right)^2$ (1); value $\frac{9}{4}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Make B equal: A:B:C $= 8:12:15$ (2); total parts $= 35$, one part $=$ ₹150 (1); A = ₹ 1200, B = ₹ 1800, C = ₹ 2250 (2).	2+1+2
34	Prism: $F = 5$, $V = 6$, $E = 9$ (2); $F + V - E = 5 + 6 - 9 = 2$ (1); pyramid: $F = 4$, $V = 4$, $E = 6$ (1); $F + V - E = 4 + 4 - 6 = 2$ (1).	2+1+1+1
35	(i) mean $= 130$ (1); (ii) rises -98, -39, 129, -25, 3; greatest in month September (2); (iii) range $= 137$ (2).	1+2+2
36	Terms $n, n+1, \ldots, n+2$ (1); sum $3n + 3$ (2); for $n = 14$ the numbers add to 45 and $3 \times 14 + 3 = 45$ (2).	1+2+2
	OR: Trapezium $= \frac{1}{2}(20 + 12) \times 9 = 144$ cm² (2); parallelogram $= 12 \times 9 = 108$ cm² (1); left over $= 36$ cm² (2).	2+1+2
37	$13824 = 2^9 \times 3^3$ (1); triplets give $\sqrt[3]{13824} = 24$ (2); check $24^3 = 13824$ (2).	1+2+2
	OR: $600000 = 6 \times 10^5$ (1); $2000 = 2 \times 10^3$ (1); product 1.2×10^9 (2); quotient 3×10^2 (1).	1+1+2+1
38	$13 = 1101_2$ (2); weights used: 1, 4, 8 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1
	OR: (i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 28^\circ$ (1); $\angle OBA = 90 - 28 = 62^\circ$ (1); $\angle ABC = 124^\circ$ (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 20	Top $= a + 3b + 3c + d$
2	**(b)** 100 cm²	$\frac{1}{2}d(h_1 + h_2) = \frac{1}{2} \times 20 \times (6 + 4)$
3	**(c)** 49	$T_n + T_{n+1} = (n+1)^2 = 49$
4	**(b)** 0.00407	Move the decimal point according to the sign of the exponent
5	**(c)** 135	Base 60: value = 60 x first + second
6	**(d)** 50°	OA = OB, so triangle OAB is isosceles
7	**(c)** 39	Sum $= 3\times$ middle number
8	**(d)** $x^2 + 7x + 10$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
9	**(a)** 10 days	Fewer or more workers: days vary inversely
10	**(b)** 20%	Discount = 500; 500/2500×100
11	**(d)** right-angled	$14^2 + 48^2 = 2500$ and $50^2 = 2500$
12	**(b)** 84	$14/60 \times 360$
13	**(c)** 6	$6(n-2)^2 = 6 \times 1$
14	**(b)** Between week 2 and week 3	Growth in each week: 4, 7, 4, 3, 6; largest is 7
15	**(c)** 90	Sum $= 9\times$ middle date
16	**(c)** 9720	$18 \times 12 \times 45$
17	**(a)** 8, 15, 17	$2m = 8$, $m^2 - 1 = 15$, $m^2 + 1 = 17$
18	**(d)** 2^{-6}	$\frac{1}{2^6} = 2^{-6}$
19	**(d)** 72	Base 20: value = 20 x first + second
20	**(b)** 110°	Angle sum is 360°

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Sum $= 16 + 61 = 77$ (1) $= 11 \times 7$ (1).	1+1
22	$b = A/l = 1200 \div 48 = 25$ m (1); perimeter $= 2(48 + 25) = 146$ m (1).	1+1
23	Take $2m = 6$, $m = 3$ (1); triplet $(6, 8, 10)$ (1).	1+1
24	$\left(\frac{3}{2}\right)^3$ (1) $= \frac{27}{8}$ (1).	1+1
25	$29 = 2 \times 14 + 1$; $14 = 2 \times 7 + 0$; $7 = 2 \times 3 + 1$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (1); 11101_2 (1).	1+1
26	Adjacent angles are supplementary: $3x + 10 + 2x - 10 = 180$ gives $x = 36$ (1); $\angle A = 118^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	XLIV = 44, LXVII = 67 (1); sum = 111 (1); CXI (1).	1+1+1
28	Diagonal AC bisects $\angle A$: $\angle OAB = 32^\circ$ (1); $\angle ABC = 180 - 64 = 116^\circ$ (1); $\angle OBA = 58^\circ$ (1).	1+1+1
29	Steps: 5321-1235=4086; 8640-0468=8172; 8721-1278=7443; 7443-3447=3996; 9963-3699=6264; 6642-2466=4176; 7641-1467=6174 (2); reaches 6174 after 7 steps (1).	2+1
30	$53 = 50 + 3$, $47 = 50 - 3$ (1); product $= 2500 - 9$ (1); $= 2491$ (1).	1+1+1
31	Man-days $= 10 \times 18 = 180$ (1); days for 15 men $= 180 \div 15$ (1); 12 days (1).	1+1+1
	OR: A = P(1+R/100)² = 12500×108/100×108/100 = ₹ 14580 (2); CI = ₹ 2080 (1).	2+1
32	$d^2 = 29^2 - 21^2 = 841 - 441 = 400$ (2); $d = 20$ m (1).	2+1
	OR: $1\% = 3.6^\circ$ (1); food 108°, rent 90° (1); education 72°, others 90° (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$83-38=45=9\times5$ (2); in general $(10a+b)-(10b+a)=9(a-b)$ (2); hence divisible by 9 (1).	2+2+1
34	$(a+b)^2=a^2+2ab+b^2$, $(a-b)^2=a^2-2ab+b^2$ (2); difference $=4ab$ (1); for $a=7$, $b=3$: $100-16=84$ and $4ab=84$ (2).	2+1+2
35	(i) speed $=240\div4=60$ km/h (1.5); (ii) $60\times7=420$ km (1.5); (iii) $360\div60=6$ hours (2).	1½+1½+2
36	Discount = ₹ 3000 (1); discounted price = ₹ 27000 (1); GST = 18% of it = ₹ 4860 (1); payable = ₹ 31860 (2).	1+1+1+2
	OR: Difference in heights $=20-12=8$ m (1); distance$^2=8^2+15^2=289$ (2); distance $=17$ m (2).	1+2+2
37	Angle per mark $=360^\circ\div60=6^\circ$ (1); angles: Maths 108°, Science 90°, English 54°, Hindi 72°, SST 36° (3); Maths percentage $=30\%$ (1).	1+3+1
	OR: Three faces: 8 (1); two faces: $12(9-2)=84$ (1); one face: $6(9-2)^2=294$ (1); none: $(9-2)^3=343$ (1); total $=8+84+294+343=729$ (1).	1+1+1+1+1
38	Mean A $=41$, mean B $=37$ (2); range A $=62$, range B $=29$ (2); more consistent: B (1).	2+2+1
	OR: $(2n+10)/2-n=n+5-n=5$ (3); for $n=7$: $(2\times7+10)/2-7=5$ (2).	3+2

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